TITLE: Implementation of a Dual-Router Network Using OSPF in One Area and Static Router in another configuring VLAN with inter-VLAN routing, DHCP, DNS, and Web Server

OBJECTIVES:

- To design a network consisting of dual routers connected using OSPF in one area and Static router in another
- To configure VLAN with inter-VLAN routing
- To set up a DHCP server.
- To configure Web and DNS servers.
- To practice troubleshooting using ping, tracert, and showing input route commands.

THEORY:

OSPF (Open Shortest Path First)

Operation: OSPF employs a link state routing mechanism, creating a comprehensive topology map of the entire network. It applies Dijkstra's algorithm to compute optimal routes and divides networks into distinct areas, with Area 0 functioning as the central backbone. Updates are shared through link state advertisements while neighbor relationships are actively monitored.

Application: Perfect for expansive enterprise environments where rapid adaption to network changes and extensive scalability are priorities. Excels in hierarchical architectures with area segmentation and performs well in topologies featuring multiple alternative paths.

Limitation: Demanding setup and diagnostic procedures, substantial processing power and memory overhead during route calculations necessitates strategic area architecture and design, challenging mastery curve for IT professionals.

Static Routing

Static route implementation refers to the manual process of creating fixed routing paths within a router's forwarding database. Network administrators explicitly specify the exact route that network traffic should traverse to reach a particular destination subnet. Functionality: The configuration is accomplished through the command When packets arrive, the router consults these predefined entries to make forwarding decisions. If connectivity is disrupted, administrators must manually intervene to reconfigure or remove the affected routing entry. Use Cases: Used in security objectives by preventing automatic route advertisement, effectively concealing network segments from potential threats.

VLAN

A Virtual Local Area Network is a technique used to divide a single physical switch into multiple logical networks. Instead of grouping devices by physical location, VLANs organize them based on function or department. This approach limits broadcast traffic, strengthens network security, and makes administration easier. Each VLAN operates as a separate broadcast domain.

VLAN Configuration:

The process of VLAN configuration begins by defining VLANs using specific identification numbers and labels. After creation, switch ports are assigned to these VLANs in access mode so connected devices become part of the selected network segment. Devices within the same VLAN can exchange data directly, whereas communication across VLANs must pass through a routing device.

Inter-VLAN Routing:

Inter-VLAN routing enables data exchange between separate VLANs through a Layer 3 device such as a router. In this setup, the Router-on-a-Stick technique is applied, where one physical router interface is logically divided into multiple sub-interfaces. Each sub-interface is linked to a VLAN and configured with 802.1Q tagging, allowing the router to manage traffic between VLANs efficiently.

DHCP(Dynamic Host Configuration Protocol)

DHCP is used to automatically provide network settings to client devices, including IP address, subnet mask, default gateway, and DNS information. This automation removes the need for manual configuration, minimizes configuration mistakes, and supports easy network expansion. In this lab setup, the server dynamically assigned IP addresses to all connected client systems.

DNS (Domain Name System):

The Domain Name System enables users to access network resources using easy-to-remember names instead of numerical IP addresses. The DNS server maintains records that associate domain names with their corresponding IP addresses and responds to client requests by providing the correct address. In this experiment, DNS allowed clients to access the server using a defined domain name.

Web Server (HTTP Service):

A web server is responsible for storing and delivering web pages to users through the HTTP protocol. When a client sends a request via a web browser, the server responds by providing the requested content, such as HTML files. In this lab, enabling the HTTP service allowed users to successfully view a webpage hosted on the server.

Integration of DHCP, DNS, and Web Services:

These network services function together to create a smooth and user-friendly network environment. DHCP handles automatic IP assignment, DNS manages name resolution, and the web server delivers requested content. Their combined operation demonstrates how centralized services enhance network efficiency, reduce configuration effort, and improve overall usability.

Network Topology

This topology shows **two branch offices** connected over a **WAN link** (dashed line) between **Router R1 Binay** and **Router R2 Binay**, both using Cisco 1841 hardware.The **left branch** has R1 connected to **Switch0**, which serves four workstations **PC0, PC1, PC4, and PC5**. The **right branch** has R2 connected to **Switch1**, supporting **PC2, PC3**, and a shared **Server0** that hosts resources accessible to both sides.Both switches are **2950-24TT** models, providing local connectivity within each branch. This simple point-to-point design allows two office locations to share centralized server resources while maintaining their own local networks.

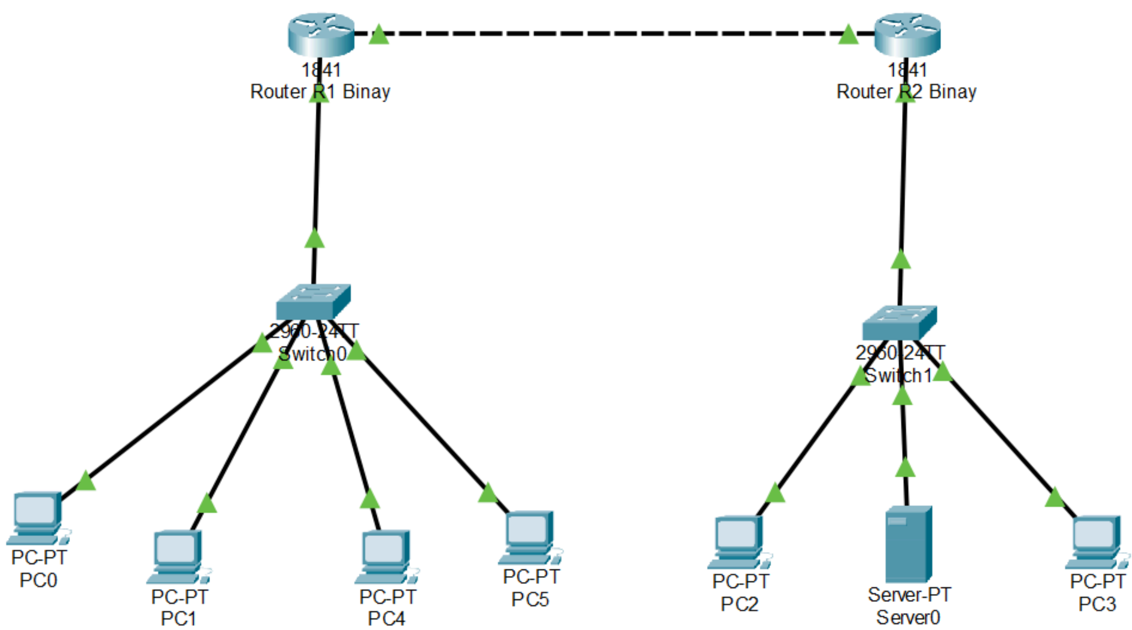


Fig: Network with dual routers using OSPF in one area and static router in another

Configuration Table:

Network devices were configured with IP addresses and subnet masks as detailed in the table below:

IP Configuration Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway	DNS Server
Router R1 Binay	FastEth0/1	192.168.1.1	255.255.255.0	N/A	N/A
Router R1 Binay	FastEth0/0	10.0.0.1	255.0.0.0	N/A	N/A
Router R2 Binay	FastEth0/1	192.168.2.1	255.255.255.0	N/A	N/A
Router R2 Binay	FastEth0/0	10.0.0.2	255.0.0.0	N/A	N/A
Server0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1	192.168.1.2
PC0	FastEthernet0	192.168.10.2	255.255.255.0	192.168.10.1	N/A
PC1	FastEthernet0	192.168.10.3	255.255.255.0	192.168.10.1	N/A
PC4	FastEthernet0	192.168.20.2	255.255.255.0	192.168.20.1	N/A
PC5	FastEthernet0	192.168.20.3	255.255.255.0	192.168.20.1	N/A
PC2	FastEthernet0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC3	FastEthernet0	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

Static Routing Configuration:

Device	Command/Route
Router R2 Binay	ip route 192.168.1.0 255.255.255.0 10.0.0.1

VLAN Assignment Table :

VLAN	Name	Interface	Network	Gateway
10	Computer	fa0/1-2	192.168.10.0/24	192.168.10.1
20	Electronics	fa0/4-5	192.168.20.0/24	192.168.20.1

PROCEDURE

1. OSPF (Configuration Commands):

Router R1 Binay

```
Router(config-if)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#
```

Router R2 Binay

```
Router(config-if)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#router ospf 1
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#
```

2. VLAN commands:

Switch Configuration

Create VLAN:

```
Switch(config)#vlan 10
Switch(config-vlan)#name Computer
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Electronics
Switch(config-vlan)#exit
Switch(config)#
```

Assign port to VLANS:

```
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access #
                                               ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
```

Configure Trunk port (switch-router)
switch(config) #interface fa0/3
#switchport mode trunk
#exit

Switch #show vlan brief

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Computer                         active    Fa0/1, Fa0/2
20   Electronics                      active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Router Configuration (Inter-VLAN Routing)

```
Router(config)#interface FastEthernet0/1.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.10, changed state to up
encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface FastEthernet0/1.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.20, changed state to up
encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

3. Procedure for DHCP, DNS and Web Server :

Server Configuration :

Step 1 : Assign Static IP to Server

1. Click on Server
2. Go to Desktop → IP Configuration
3. Configure the following:
    - IP Address: 192.168.1.2
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.1.1
    - DNS Server: 192.168.1.2

DHCP Configuration :

Step 2 : Enable DHCP Service

1. Go to Server → Services → DHCP
2. Turn DHCP: ON

Step 3 : Create DHCP Pool

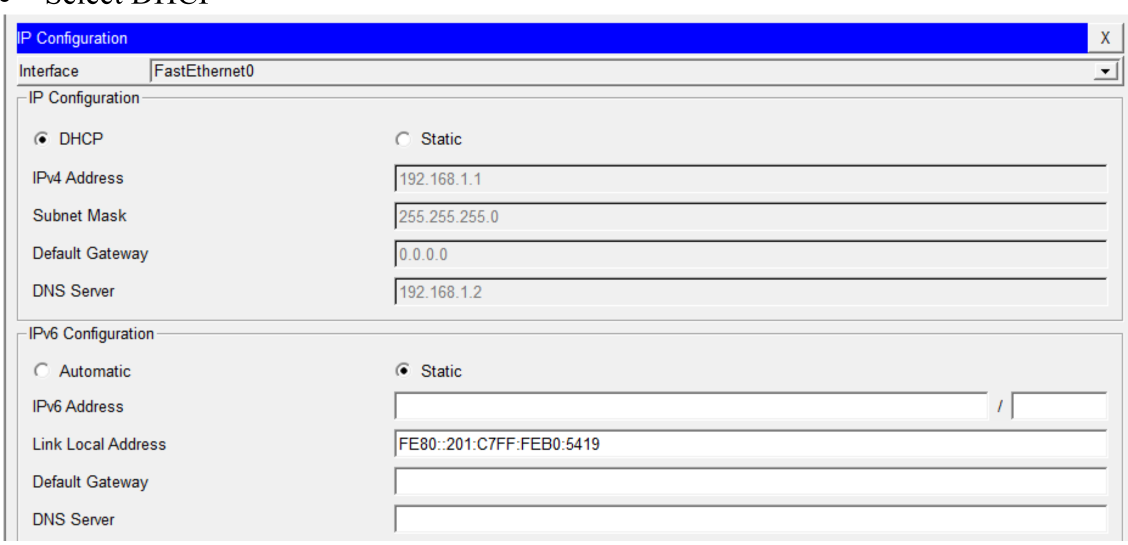
Field	**Value**
Pool Name	LANpool
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.0
Subnet Mask	255.255.255.0
Maximum Users	50

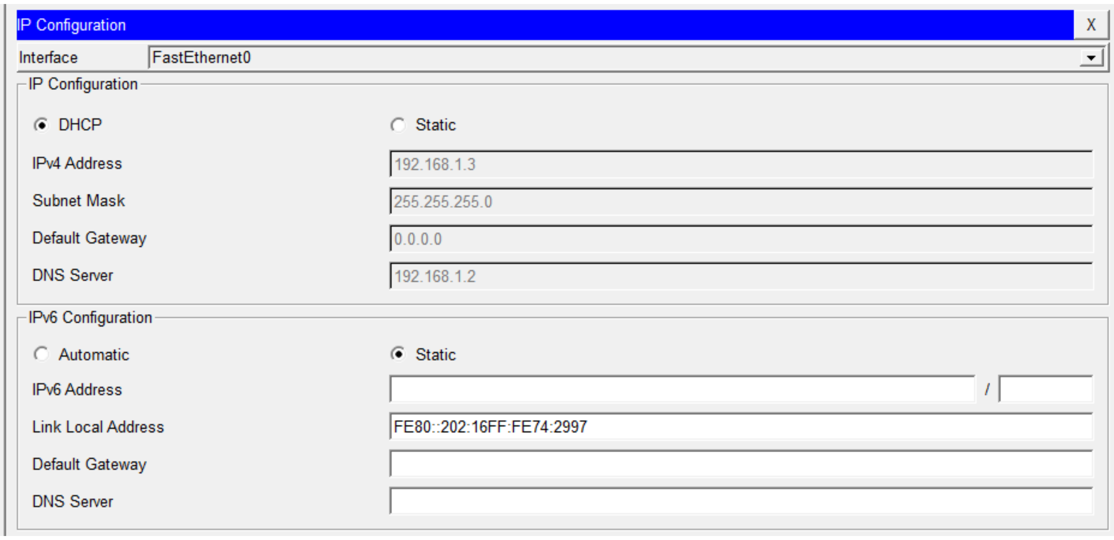
Click Add

Verification :

On each PC:

- Go to Desktop → IP Configuration
- Select DHCP






```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=7ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 1ms
```

Fig: DHCP ping

DNS Server Configuration :

Step 4 : Enable DNS Service

1. Go to Server → Services → DNS
2. Turn DNS: ON

Step 5 : Add DNS Records

- Name: www.hcoe.com
- Type: A record
- Address:

192.168.1.2

Click Add

Verification :
On each PC:

- Open Command Prompt
- Ping: www.binayhcoe.com

```
C:\>ping www.binayhcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fig: DNS ping on PC2

```
C:\>ping www.binayhcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fig: DNS ping on PC3

Web Server Configuration :
Step 6 : Enable HTTP Service

1. Go to Server → Services → HTTP
2. Turn HTTP: ON
3. Edit index.html (optional)

Verification :
On PC:

- Go to Desktop → Browser
- Type: http://www.binayhcoe.com

Troubleshooting:

Ping:

OSPF ping: PC0 - PC2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 4ms
```

Fig: OSPF ping

Static routing ping:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Static Routing ping

Tracting:

OSPF:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.2.2

Tracing route to 192.168.2.2 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.1
  2   0 ms      0 ms      0 ms      10.0.0.2
  3   *         0 ms      1 ms      192.168.2.2

Trace complete.
```

Fig: tracting OSPF

Static routing:

```
C:\>tracert 192.168.2.3

Tracing route to 192.168.2.3 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.2.3

Trace complete.
```

Fig: tracting static routing

DNS server:

```
C:\>tracert 192.168.1.3

Tracing route to 192.168.1.3 over a maximum of 30 hops:

  1   0 ms      1 ms      1 ms      192.168.1.3

Trace complete.
```

Fig: tracting DNS server

Tracting VLAN: from PC0 vlan 10 computer PC4

```
C:\>tracert 192.168.20.2

Tracing route to 192.168.20.2 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.1
  2   0 ms      0 ms      0 ms      192.168.20.2

Trace complete.
```

Tracting VLAN from PC4 vlan 20 electronics to PC0

```
C:\>tracert 192.168.10.2

Tracing route to 192.168.10.2 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.20.1
  2   1 ms      0 ms      0 ms      192.168.10.2

Trace complete.
```

Discussion

This project demonstrated the implementation of a dual router network using OSPF and static routing in Cisco Packet Tracer. OSPF between R1 and R2 in Area 0 enabled dynamic route exchange, with traceroute confirming packets traversed the correct 3-hop path through the WAN link. Static routing on R2 for the 192.168.1.0/24 network provided a simple, controlled method for return traffic, and the initial packet loss in ping tests was expected due to ARP resolution on the first request.

VLAN segmentation on switch0 separated the network into VLAN 10 (Computer) and VLAN 20 (Electronics). The router on a stick technique with 802.1Q sub-interfaces successfully enabled inter-VLAN communication, as verified by traceroute showing traffic routing through the gateway. DHCP, DNS and the web server worked together seamlessly clients received IPs automatically, resolved www.binayhcoe.com by name, and accessed the hosted webpage through browser.

Conclusion

This project successfully achieved all the stated objectives. A dual-router network was configured using OSPF in one area and Static routing in another for controlled traffic flow, with VLAN segmentation and inter-VLAN routing implemented using the router on a stick technique. DHCP, DNS and HTTP services were integrated on a centralized server, demonstrating how these services enhance network efficiency and usability.

The troubleshooting exercises using ping and tracert confirmed end-to-end connectivity across all network segments and VLANs. Overall, this lab provided valuable hands-on experience in enterprise network design, reinforcing the importance of proper IP planning, routing protocol selection, and systematic troubleshooting.